

Lanka Traffic Fine System

MongoDB Database Schema — SIMPLIFIED for University Project · v1.1

SIMPLIFIED VERSION — 4 CHANGES FROM ORIGINAL

- Refresh tokens removed** — single JWT access token (24 h expiry) replaces the rotation system.
- Payment gateway simplified** — mock endpoint returns SUCCESS immediately; no PayHere webhooks.
- Districts hardcoded in frontend** — no backend CRUD for districts; seeded once at startup.
- Geospatial index removed** — location fields kept as optional strings; no 2dsphere index.

Database Overview

Database Engine	MongoDB 7.x (Atlas / self-hosted)
ODM	Mongoose 8.x
Naming Convention	snake_case collection names, camelCase field names
ID Strategy	MongoDB ObjectId (_id) — 24-char hex string
Timestamps	createdAt / updatedAt auto-managed by Mongoose { timestamps: true }
Soft Deletes	isActive: Boolean (never hard-delete users or categories)
Transactions	Multi-document transactions used for Payment + Fine status update atomically
Auth Tokens	Single JWT access token — 24 h expiry. No refresh token collection.
Environment	Separate databases: trafficfines_dev, trafficfines_test, trafficfines_prod

Collections at a Glance

Collection	Purpose	Est. Rows/Year	Key Relationships
users	Drivers, Officers, Admins	~50,000	→ districts, fines
districts	Sri Lanka's 25 admin districts (seeded, read-only)	25 (static)	← users, fines
fine_categories	Violation types with fixed fine amounts	~50 (semi-static)	← fines
fines	Issued traffic fines — core entity	~500,000	→ users, districts, fine_categories
payments	Payment records (mock gateway)	~400,000	→ fines (1:1)
notifications	SMS dispatch log to officers	~400,000	→ fines, users

■ refresh_tokens collection has been REMOVED. Authentication uses a single 24-hour JWT instead.

① users — Drivers, Officers and Admins

Field	Type	Req	Default	Key	Notes / Constraints
_id	ObjectId	✓	—	PK	Auto-generated MongoDB ObjectId
fullName	String	✓	—	—	Min 2, Max 100 chars
email	String	✓	—	UNQ	Lowercase, trimmed. Unique per system.
phone	String	✓	—	IDX	Sri Lankan format: 07XXXXXXXX or +947XXXXXXXX

passwordHash	String	✓	–	–	bcrypt hash (rounds=12). Never stored plain.
role	String (enum)	✓	–	IDX	enum: DRIVER OFFICER ADMIN
licenseNo	String	–	–	IDX	Driver license. Required when role=DRIVER
badgeNo	String	–	–	UNQ	Badge number. Required when role=OFFICER.
districtId	ObjectId ref	–	–	FK	Ref → districts._id. Required when role=OFFICER
isActive	Boolean	✓	true	–	Soft disable without deletion
lastLoginAt	Date	–	–	–	Updated on every successful login
createdAt	Date	✓	now	–	Auto by Mongoose timestamps
updatedAt	Date	✓	now	–	Auto by Mongoose timestamps

Validation: email regex · phone regex `/(+94|0)7[0-9]{8}$/` · password min 8 chars with uppercase, lowercase, digit · role strictly one of DRIVER/OFFICER/ADMIN

② districts — Sri Lanka's 25 Administrative Districts (Seeded / Read-Only)

■ **Simplified** Districts are seeded once at startup from a static list. No backend CRUD endpoints are needed. Frontend applications hardcode the district list. The collection exists only for FK references.

Field	Type	Req	Default	Key	Notes
_id	ObjectId	✓	–	PK	Auto ObjectId
name	String	✓	–	UNQ	English district name e.g. 'Colombo'
code	String	✓	–	UNQ	2–3 char code e.g. 'CMB', 'KDY'
province	String	✓	–	IDX	Western Central Southern Northern Eastern etc.
isActive	Boolean	✓	true	–	All 25 districts active by default
createdAt	Date	✓	now	–	Mongoose auto
updatedAt	Date	✓	now	–	Mongoose auto

Code	District	Province
CMB	Colombo	Western
GMP	Gampaha	Western
KLT	Kalutara	Western
KDY	Kandy	Central
MTL	Matale	Central
NUW	Nuwara Eliya	Central
GLG	Galle	Southern
MTR	Matara	Southern
HMB	Hambantota	Southern
JFN	Jaffna	Northern
KLN	Kilinochchi	Northern
MNR	Mannar	Northern
VVN	Vavuniya	Northern
MLT	Mullaitivu	Northern
TRC	Trincomalee	Eastern
BTL	Batticaloa	Eastern
AMP	Ampara	Eastern
KUR	Kurunegala	North Western

PUT	Puttalam	North Western
ANU	Anuradhapura	North Central
PLN	Polonnaruwa	North Central
BDL	Badulla	Uva
MON	Monaragala	Uva
RAT	Ratnapura	Sabaragamuwa
KEG	Kegalle	Sabaragamuwa

③ fine_categories — Traffic Violation Types with Fixed Penalty Amounts

Field	Type	Req	Default	Key	Notes
_id	ObjectId	✓	—	PK	Auto ObjectId
code	String	✓	—	UNQ	Short unique code e.g. SP-01, DUI-01
name	String	✓	—	—	Human-readable category name
amount	Number	✓	—	—	Fine amount in LKR. Min: 500. Decimal128.
demeritPoints	Number	✓	0	—	Points against license. 0–10 range.
description	String	—	—	—	Full legal description of the violation
legalRef	String	—	—	—	Motor Traffic Act section reference
isActive	Boolean	✓	true	—	Inactive categories cannot be selected for new fines
createdAt	Date	✓	now	—	Mongoose auto
updatedAt	Date	✓	now	—	Mongoose auto

Code	Category Name	Amount (LKR)	Demerit Pts	Legal Ref
SP-01	Speeding 1–20 km/h over	1,500	2	S.138(1)
SP-02	Speeding 21–40 km/h over	3,000	4	S.138(1)
SP-03	Speeding 41+ km/h over	6,000	6	S.138(2)
DUI-01	Driving under influence	25,000	10	S.151
SBT-01	Not wearing seat belt	1,000	1	S.148(1)
MOB-01	Using mobile while driving	2,500	3	S.149
RLS-01	Running red signal	3,000	4	S.135
OVT-01	Illegal overtaking	2,000	3	S.136
NL-01	Driving without valid license	5,000	5	S.122
INS-01	No valid insurance	10,000	4	MV Ins. Act S.3
PKG-01	Parking in no-parking zone	1,500	1	S.146
NHW-01	Wrong way on highway	5,000	5	S.137

④ fines — Core Entity: Every Issued Traffic Fine

■ **Simplified** GPS location fields (lat/lng) are stored as optional strings only. The 2dsphere geospatial index has been REMOVED. No location tracking is required.

Field	Type	Req	Default	Key	Notes
_id	ObjectId	✓	—	PK	Auto ObjectId
referenceNo	String	✓	—	UNQ	Pattern: TF-YYYYMMDD-NNNNNN. Auto-generated server-side.
vehicleNo	String	✓	—	IDX	Uppercase, trimmed. e.g. CAB-1234

driverName	String	–	–	–	Optional at issue time
driverLicense	String	–	–	IDX	Driver license number (optional at issue)
officerId	ObjectId ref	✓	–	FK	Ref → users._id (role=OFFICER)
categoryId	ObjectId ref	✓	–	FK	Ref → fine_categories._id
districtId	ObjectId ref	✓	–	FK	Ref → districts._id
amount	Decimal128	✓	–	–	Copied from category at issue time. Immutable.
status	String (enum)	✓	–	IDX	PENDING PAID OVERDUE CANCELLED
location.address	String	–	–	–	Optional human-readable address string (no GPS required)
notes	String	–	–	–	Officer's free-text notes. Max 500 chars.
issuedAt	Date	✓	now	–	Timestamp of fine issuance
dueDate	Date	✓	–	–	issuedAt + 30 days. After this → OVERDUE.
paidAt	Date	–	–	–	Set when status changes to PAID
cancelledAt	Date	–	–	–	Set when status changes to CANCELLED
cancelledBy	ObjectId ref	–	–	FK	Ref → users._id (who cancelled)
createdAt	Date	✓	now	–	Mongoose auto
updatedAt	Date	✓	now	–	Mongoose auto

Status Transition Rules: PENDING → PAID (on mock payment success) · PENDING → OVERDUE (cron: dueDate < now) · PENDING → CANCELLED (officer/admin) · OVERDUE → PAID (late payment) · PAID / CANCELLED are terminal states.

Reference Number Generation

```
// utils/referenceGenerator.js
const generateReferenceNo = async () => {
  const date = new Date();
  const dateStr = date.toISOString().slice(0,10).replace(/-/g,''); // 20260513
  const count = await Fine.countDocuments({ createdAt: { $gte: startOfDay(date) } });
  const seq = String(count + 1).padStart(6, '0'); // 000123
  return `TF-${dateStr}-${seq}`; // TF-20260513-000123
};
```

⑤ payments — Payment Records (Simplified Mock Gateway)

■ Simplified

Real PayHere / payment gateway integration has been REMOVED. The backend exposes a single mock endpoint that accepts payment details and immediately returns SUCCESS. No webhook handler, no HMAC signature validation, no paymentIntentId handshake required.

Field	Type	Req	Default	Key	Notes
_id	ObjectId	✓	–	PK	Auto ObjectId
fineId	ObjectId ref	✓	–	FK, UNQ	Ref → fines._id. Unique — one payment per fine.
receiptNo	String	–	–	UNQ	Pattern: RCP-YYYYMMDD-NNNNNN. Set on SUCCESS.
payerName	String	✓	–	–	Name of person making payment

payerPhone	String	✓	–	–	Phone for SMS receipt
payerEmail	String	–	–	–	Optional email for digital receipt
amount	Decimal128	✓	–	–	Must match fines.amount. Validated server-side.
currency	String	✓	LKR	–	ISO 4217 currency code
paymentMethod	String (enum)	✓	–	–	CARD ONLINE_BANKING MOBILE_WALLET
cardNumber	String	–	–	–	Last 4 digits only (never store full PAN)
status	String (enum)	✓	–	IDX	PENDING SUCCESS FAILED
paidAt	Date	–	–	–	Timestamp of successful mock payment
createdAt	Date	✓	now	–	Mongoose auto
updatedAt	Date	✓	now	–	Mongoose auto

Atomic Transaction Pattern (still required for data integrity)

```
// On mock payment confirm – use MongoDB transaction
const session = await mongoose.startSession();
session.startTransaction();
try {
  await Payment.findByIdAndUpdate(paymentId, { status:'SUCCESS', paidAt, receiptNo }, { session });
  await Fine.findByIdAndUpdate(fineId, { status:'PAID', paidAt }, { session });
  await session.commitTransaction();
  await smsService.sendToOfficer(fine, payment); // outside transaction (non-critical)
} catch (err) {
  await session.abortTransaction();
  throw err;
} finally { session.endSession(); }
```

⑥ notifications — SMS Dispatch Log

Field	Type	Req	Default	Key	Notes
_id	ObjectId	✓	–	PK	Auto ObjectId
fineId	ObjectId ref	✓	–	FK	Ref → fines._id
paymentId	ObjectId ref	✓	–	FK	Ref → payments._id
officerId	ObjectId ref	✓	–	FK	Ref → users._id (the receiving officer)
recipientPhone	String	✓	–	–	Officer's phone number at time of send
message	String	✓	–	–	Full SMS text as sent. Max 160 chars.
type	String (enum)	✓	SMS	–	SMS (extensible to PUSH / EMAIL)
status	String (enum)	✓	–	IDX	PENDING SENT DELIVERED FAILED
gateway	String	✓	–	–	NOTIFY_LK DIALOG MOBITEL
gatewayRef	String	–	–	–	Gateway message ID
retryCount	Number	✓	0	–	Number of retry attempts. Max 3.
sentAt	Date	–	–	–	When SMS was dispatched to gateway
createdAt	Date	✓	now	–	Mongoose auto
updatedAt	Date	✓	now	–	Mongoose auto

SMS Template: Fine {referenceNo} for vehicle {vehicleNo} has been PAID (LKR {amount}) on {date}. Receipt: {receiptNo}. Driver may collect license. – SL Traffic Fine System

Aggregation Pipelines — Admin Analytics

1. District-wise Revenue Summary

```
db.fines.aggregate([
  { $match: { status: 'PAID', paidAt: { $gte: fromDate, $lte: toDate } } },
  { $group: { _id: '$districtId', totalIssued: { $sum: 1 },
    revenue: { $sum: { $toDouble: '$amount' } } } },
  { $lookup: { from: 'districts', localField: '_id', foreignField: '_id', as: 'district' } },
  { $unwind: '$district' },
  { $project: { districtName: '$district.name', province: '$district.province',
    totalIssued: 1, revenue: 1 } },
  { $sort: { revenue: -1 } }
])
```

2. Daily / Monthly Trend

```
db.payments.aggregate([
  { $match: { status: 'SUCCESS', paidAt: { $gte: fromDate, $lte: toDate } } },
  { $group: { _id: { $dateTrunc: { date: '$paidAt', unit: granularity,
    timezone: 'Asia/Colombo' } },
    finesPaid: { $sum: 1 }, revenue: { $sum: { $toDouble: '$amount' } } } },
  { $sort: { _id: 1 } },
  { $project: { date: '$_id', finesPaid: 1, revenue: 1, _id: 0 } }
])
```

Environment Configuration

```
# MongoDB
MONGO_URI=mongodb+srv://<user>:<pass>@cluster.mongodb.net/trafficfines_prod
MONGO_DB_NAME=trafficfines_prod

# JWT – single token, 24-hour expiry (no refresh token)
JWT_SECRET=<32+ random bytes hex>
JWT_EXPIRES=24h

# SMS Gateway (Notify.lk)
SMS_GATEWAY=NOTIFY_LK
SMS_API_KEY=<your_key>
SMS_SENDER_ID=TrafficFine

# App
NODE_ENV=production
PORT=5000
FRONTEND_URL=https://trafficfines.lk
ADMIN_URL=https://admin.trafficfines.lk
```